

Navigating High-Dimensional Search Spaces: A Comparative Study of Stochastic Strategies for Complex Optimization Problems

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Abstract—The Grey Wolf Optimizer (GWO) is a population-based metaheuristic that models the leadership hierarchy and cooperative hunting behavior of grey wolves to solve continuous, black-box optimization problems [1]. Since its introduction, GWO has been widely adopted in engineering applications, yet independent, statistically grounded replications that place it against a well-established baseline under matched, controlled conditions remain useful for practitioners deciding which metaheuristic to reach for. This paper reports a ground-up implementation of GWO and a baseline inertia-weight Particle Swarm Optimizer (PSO) [2], [3], both benchmarked on four standard continuous test functions (Sphere, Shifted Sphere, Rastrigin, and Rosenbrock) spanning unimodal, multimodal, separable, and non-separable landscapes, across three problem dimensionalities (2, 10, 30) and 25 independent trials per configuration (600 runs total). We report the best, worst, mean, median, standard deviation fitness alongside, success rates, function-evaluation budgets, and paired with Wilcoxon signed-rank tests for statistical significance. GWO outperformed the PSO baseline with statistical significance ($p < 0.05$) in 6 of 12 configurations. PSO had a significant edge in 2 low-dimensional configurations and the remaining 4 did not show significant differences. The GWO's advantage grew with problem dimensionality, consistent with its adaptive exploration-exploitation balance through the linearly decreasing coefficient a . Both algorithms substantially degraded on high-dimensional Rastrigin and Rosenbrock, illustrating that no method escapes the curse of dimensionality on genuinely hard landscapes. We formulate this work honestly as an implementation-and-replication study with a genuine comparative-empirical contribution, not as a novel algorithmic proposal.

Index Terms—Grey Wolf Optimizer, Particle Swarm Optimization, metaheuristics, swarm intelligence, benchmark functions, empirical algorithm comparison

I. INTRODUCTION

CONTINUOUS, derivative-free (black-box) optimization is central to engineering design, where objective functions are frequently non-convex, non-differentiable, or expensive to evaluate via simulation. Nature-inspired metaheuristics, algorithms that borrow search strategies from biological, physical, or social processes, have become a standard tool for such problems because they do not require gradient information and degrade gracefully in multimodal landscapes [4]. Among these, the Grey Wolf Optimizer (GWO), introduced by [1], models the social hierarchy of a wolf pack (alpha, beta, delta, omega) and an encircle-and-attack hunting strategy to balance global exploration and local exploitation. GWO has attracted considerable continuing research [5] catalogue dozens of hybridizations and application-domain deployments in their review, because of its structural simplicity (few tunable

hyperparameters, no crossover or mutation machinery) and competitive empirical performance in the original paper's 29-function benchmark. This project began as master's coursework for *Optimization Methods in Engineering Applications* (April 2025) and originally consisted of (i) a from-scratch implementation of GWO, (ii) qualitative and 3-D visualizations of four standard test functions, and (iii) exploratory single-run and small-sample demonstrations of GWO's convergence behavior. That version, while pedagogically sound, had three limitations common to coursework level implementations that we address in this paper: the absence of a baseline comparator, the absence of statistical testing, and no explicit accounting of computational budget when comparing runs.

II. RESEARCH PROBLEM

We formulate the research problem as follows:

Under matched population size, iteration budget, and function-evaluation count, does a faithful implementation of the Grey Wolf Optimizer produce statistically better final solution quality than a standard inertia-weight Particle Swarm Optimizer, across benchmark functions of varying modality and separability, and how does any advantage scale with problem dimensionality?

A. Contributions

We emphasize that this paper does not propose a new algorithm, a new hybrid variant, or a new benchmark. Its contributions are:

- An independent re-implementation of GWO [1], reproducing the population dynamics, hierarchy-update, and position-update equations exactly as specified in the original paper.
- A matched-budget PSO baseline [2], [3], implemented independently but under an identical experimental protocol.
- A statistically grounded comparative benchmarking study across 12 function-dimension configurations with 25 independent trials each (600 total runs), including paired Wilcoxon signed-rank testing, success-rate analysis, and dimensional-scalability analysis.
- A transparent discussion of where GWO's advantage holds, where it disappears, and where both algorithms fail, contrasted with the No Free Lunch theorem [6].

TABLE I: Benchmark functions.

Function	Formula	Domain	Global optimum	Class
Sphere	$\sum x_i^2$	$[-5.12, 5.12]^d$	$f(\mathbf{0}) = 0$	Unimodal, convex, separable
Shifted Sphere	$\sum (x_i - \sqrt{2})^2$	$[-5.12, 5.12]^d$	$f(\sqrt{2}\mathbf{1}) = 0$	Unimodal, shifted optimum
Rastrigin	$10d + \sum [x_i^2 - 10 \cos(2\pi x_i)]$	$[-5.12, 5.12]^d$	$f(\mathbf{0}) = 0$	Multimodal, separable
Rosenbrock	$\sum [100(x_{i+1} - x_i^2)^2 + (1 - x_i)^2]$	$[-2.048, 2.048]^d$	$f(\mathbf{1}) = 0$	Non-convex, non-separable

This is, in short, a rigorous **replication and comparative benchmarking study**, distinct from (and not overstated as) a novel-algorithm contribution.

III. RELATED WORK

Swarm and hierarchy-based metaheuristics: Particle Swarm Optimization [2] was among the first population-based metaheuristics to model social information-sharing. [3] introduced the inertia-weight variant used here, which linearly anneals the exploration-exploitation trade-off over the search, structurally analogous to GWO’s linearly decreasing coefficient a , making PSO a natural comparator for GWO. Grey Wolf Optimizer and variants: GWO [1] was originally benchmarked on 29 functions against PSO, Gravitational Search Algorithm, and Differential Evolution. [5] review the subsequent literature, documenting binary, multi-objective, chaotic and hybrid GWO variants, and note that most application papers report single-run or small-sample comparisons without formal significance testing, a gap this paper addresses for the base algorithm. Benchmark functions: The functions used here are drawn from the standard continuous-optimization test-function literature classified by [7], chosen to span the modality $\times$ separability design space. No Free Lunch Theorems: [6]’s No Free Lunch theorems establish that, averaged over all possible objective functions, no optimizer outperforms any other. This motivates our explicit framing. Any comparative advantage reported here is a statement about performance on this specific, well categorized function class.

IV. PROBLEM DEFINITION

We consider the unconstrained, continuous, black-box minimization problem

$$\min_{\mathbf{x} \in [l, u]^d} f(\mathbf{x}) \quad (1)$$

where f is only evaluable pointwise, d is the problem dimension, and $[l, u]$ defines a box-constrained search domain. We study four instances of f (Table I) that jointly vary modality (unimodal vs. multimodal) and separability (separable vs. non-separable).

Each function is evaluated at $d \in \{2, 10, 30\}$, giving 12 function-dimension configurations. We define success on a run as reaching the best-so-far fitness below 10^{-3} .

V. METHODS

A. Grey Wolf Optimizer

GWO maintains a population of candidate solutions n (“wolves”) in R^d . At each iteration, the three best solutions are labelled alpha (α), beta (β), and delta (δ); all remaining

“omega” wolves update their position by averaging three candidate positions, each computed by extrapolating or interpolating toward one of the three leaders:

$$D_{\text{leader}} = |C \cdot X_{\text{leader}} - X| \quad (2)$$

$$X_{\text{cand}} = X_{\text{leader}} - A \cdot D_{\text{leader}} \quad (3)$$

for leader $\in \{\alpha, \beta, \delta\}$, where $A = 2ar_1 - a$ and $C = 2r_2$, with $r_1, r_2 \sim U(0, 1)$ drawn independently per dimension and per leader. The coefficient a decreases linearly from 2 to 0 over the run, transitioning the population from exploration ($|A| > 1$) to exploitation ($|A| < 1$). Each omega wolf’s new position is the coordinate-wise average of the three leader-guided candidates, clipped to the box constraints and re-evaluated. This is a direct, unmodified re-implementation of [1]’s update rule; we introduced no structural changes or enhancements from the variant literature [5], so as to evaluate the base algorithm as originally proposed.

B. Particle Swarm Optimization (comparator)

As a matched baseline we implemented the inertia-weight PSO of [3], building on [2]. Each particle i maintains a position $\mathbf{x}_i$ and velocity $\mathbf{v}_i$, updated as

$$\mathbf{v}_i \leftarrow w\mathbf{v}_i + c_1r_1(\mathbf{p}_i - \mathbf{x}_i) + c_2r_2(\mathbf{g} - \mathbf{x}_i) \quad (4)$$

$$\mathbf{x}_i \leftarrow \text{clip}(\mathbf{x}_i + \mathbf{v}_i, l, u) \quad (5)$$

where $\mathbf{p}_i$ is particle i ’s personal best, $\mathbf{g}$ is the swarm’s global best, $r_1, r_2 \sim U(0, 1)$, $c_1 = c_2 = 2.0$, and the inertia weight w decreases linearly from 0.9 to 0.4 over the run, deliberately mirroring GWO’s linear exploration-to-exploitation anneal.

C. Experimental procedure

Both algorithms were run with population size 30, 200 iterations, and identical random seeds per trial (seed = run index, 0–24), so both algorithms see the same initial population draw within a trial. Each of the 12 configurations was run for 25 independent trials per algorithm (600 optimization runs total). We recorded best, worst, mean, median, and standard deviation of final best fitness, success rate, mean wall-clock time, and mean function evaluations. We then tested for significance using the paired Wilcoxon signed-rank test. All code used Python 3, NumPy, SciPy, and Matplotlib.

VI. EXPERIMENTS

We organized the empirical study around three questions: (Q1) solution quality and significance; (Q2) reliability (success rate, variance); (Q3) scalability with dimension. Figure 1 shows the four benchmark landscapes in two dimensions.

Figure 1. Two-dimensional landscapes of the benchmark test functions

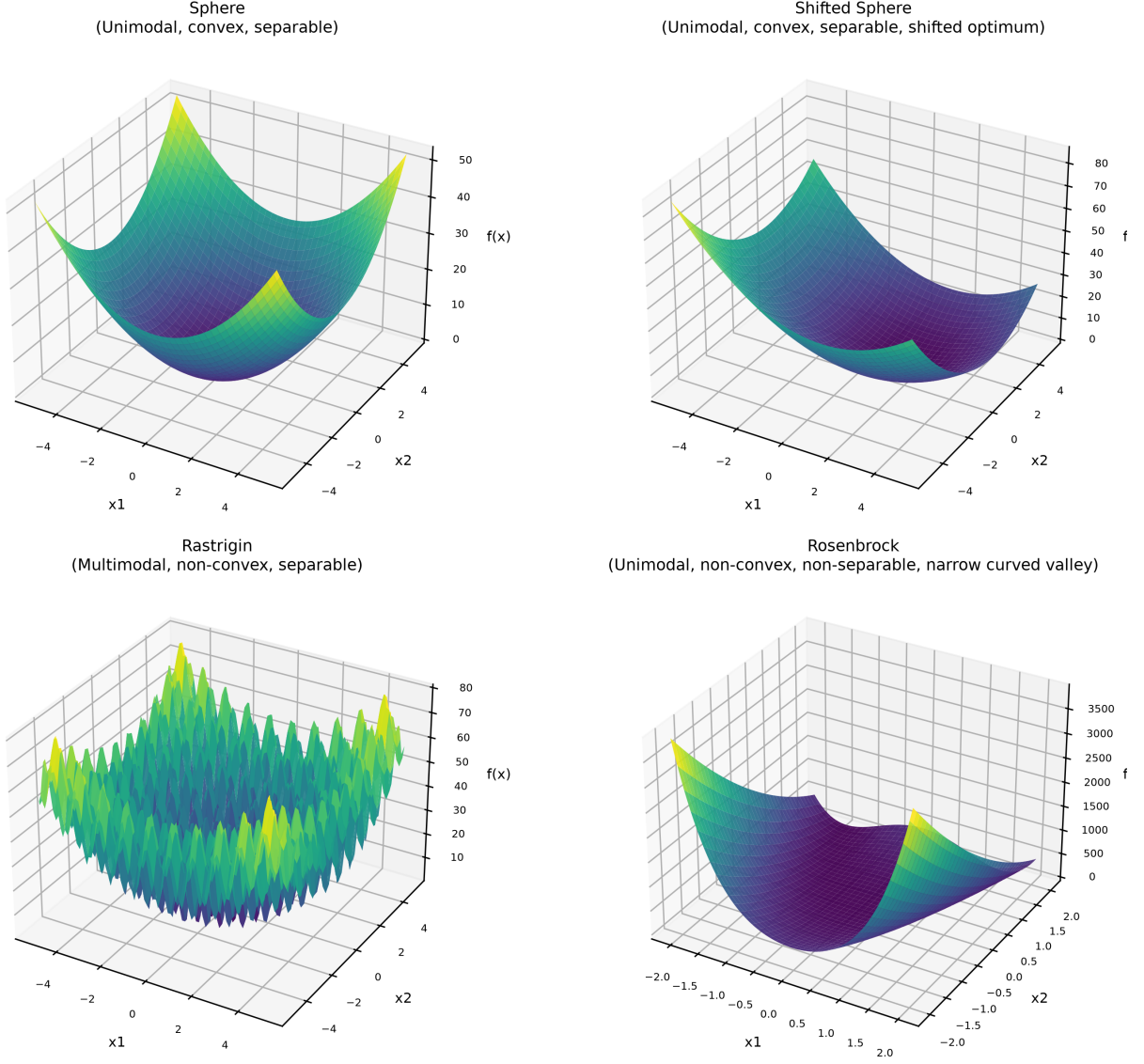


Fig. 1: Two-dimensional landscapes of the four benchmark test functions.

VII. RESULTS

Table II summarizes all 12 configurations. Figure 2 shows mean convergence trajectories (± 1 s.d. in 25 runs), Figure 3 compares the final fitness distributions in 10D, Figure 4 compares the success rates and Figure 5 shows the scaling of the mean final fitness with dimension.

TABLE II: Summary of results across all 12 function–dimension configurations (25 runs each). Bold indicates the statistically better algorithm (Wilcoxon $p < 0.05$); "–" indicates no significant difference.

Function	Dim	GWO mean ($\pm$ s.d.)	PSO mean ($\pm$ s.d.)	GWO succ.%	PSO succ.%	Wilcoxon p	Better
Sphere	2	2.35e-91 ($\pm$ 9.70e-91)	3.72e-19 ($\pm$ 6.39e-19)	100	100	1.00	–
Sphere	10	1.32e-25 ($\pm$ 2.86e-25)	6.25e-05 ($\pm$ 9.32e-05)	100	100	5.96e-08	GWO
Sphere	30	2.82e-12 ($\pm$ 2.86e-12)	7.74 ($\pm$ 9.37)	100	0	5.96e-08	GWO
Shifted Sphere	2	3.60e-07 ($\pm$ 4.29e-07)	7.22e-19 ($\pm$ 2.14e-18)	100	100	5.96e-08	PSO
Shifted Sphere	10	6.51e-05 ($\pm$ 2.30e-05)	7.50e-05 ($\pm$ 1.02e-04)	100	100	0.44	–
Shifted Sphere	30	3.12 ($\pm$ 2.18)	25.94 ($\pm$ 15.59)	0	0	2.56e-06	GWO
Rastrigin	2	0.00 ($\pm$ 0.00)	1.14e-14 ($\pm$ 3.29e-14)	100	100	1.00	–
Rastrigin	10	5.18 ($\pm$ 8.14)	19.64 ($\pm$ 7.72)	36	0	2.66e-05	GWO
Rastrigin	30	33.40 ($\pm$ 27.51)	249.55 ($\pm$ 40.05)	0	0	5.96e-08	GWO
Rosenbrock	2	7.47e-06 ($\pm$ 1.86e-05)	8.10e-07 ($\pm$ 1.68e-06)	100	100	3.76e-04	PSO
Rosenbrock	10	6.94 ($\pm$ 0.65)	28.03 ($\pm$ 84.06)	0	0	0.17	–
Rosenbrock	30	27.34 ($\pm$ 0.83)	447.65 ($\pm$ 343.10)	0	0	5.96e-08	GWO

Across the 12 configurations, GWO achieved a statistically significant advantage in 6, PSO in 2 (both at low dimension: Shifted Sphere 2D and Rosenbrock 2D) and no significant difference in the remaining 4 (Sphere 2D, Shifted Sphere 10D, Rastrigin 2D, Rosenbrock 10D). GWO’s advantage was concentrated at higher dimensionality: at 10D and 30D, GWO’s mean final fitness was one to two orders of magnitude better than PSO’s on all four functions. Success rates follow the same pattern. The PSO success rate collapsed at 10D/30D on every function, while GWO retained partial success (e.g., 36% on 10D Rastrigin) where PSO achieved 0%. Neither algorithm reached the success threshold on Rastrigin-30D, Rosenbrock-10D/30D, or Shifted-Sphere-30D, demonstrating that GWO’s relative advantage does not translate into absolute problem-solving capability on genuinely hard, high-dimensional landscapes within the tested budget.

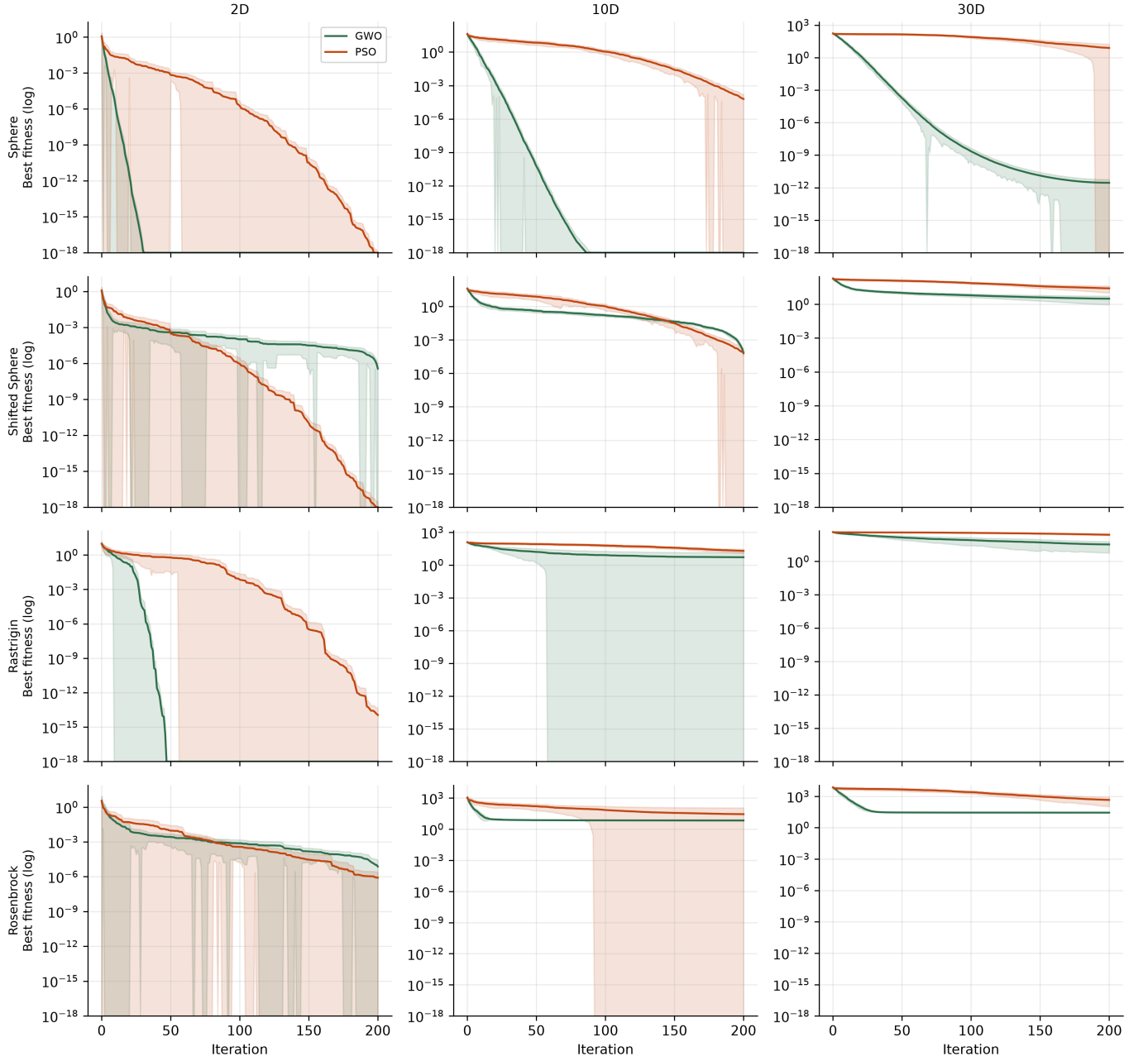
Figure 2. Convergence trajectories (mean $\pm$ 1 s.d. over 25 independent runs): GWO vs PSOFig. 2: Convergence trajectories (mean $\pm$ 1 s.d. over 25 independent runs): GWO vs. PSO.

Figure 3. Final solution quality at 10D (25 independent runs)

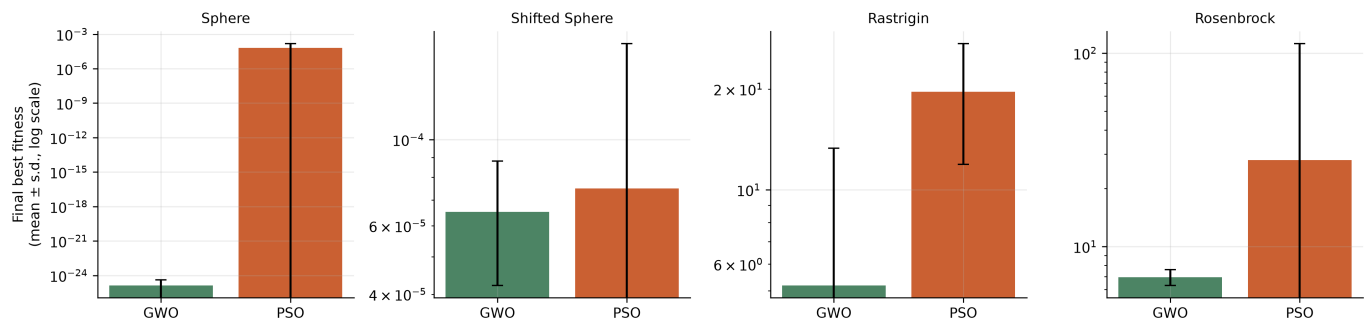


Fig. 3: Final solution quality at 10D (25 independent runs).

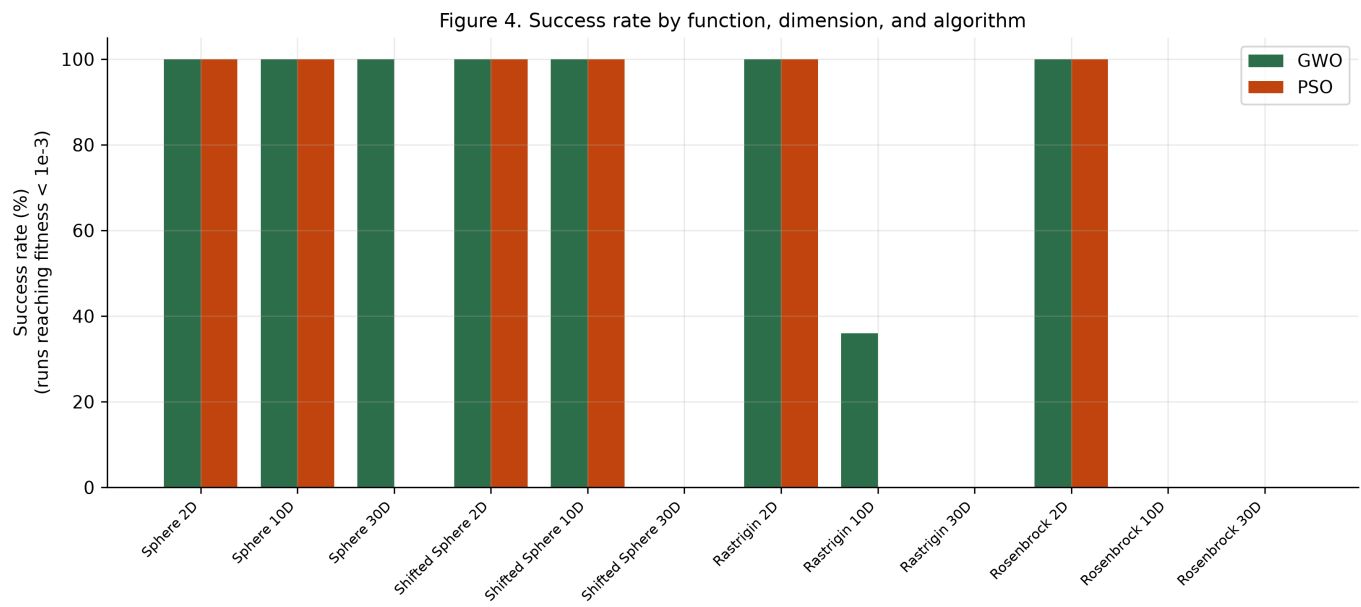


Fig. 4: Success rate by function, dimension, and algorithm.

Figure 5. Scalability of solution quality with problem dimension

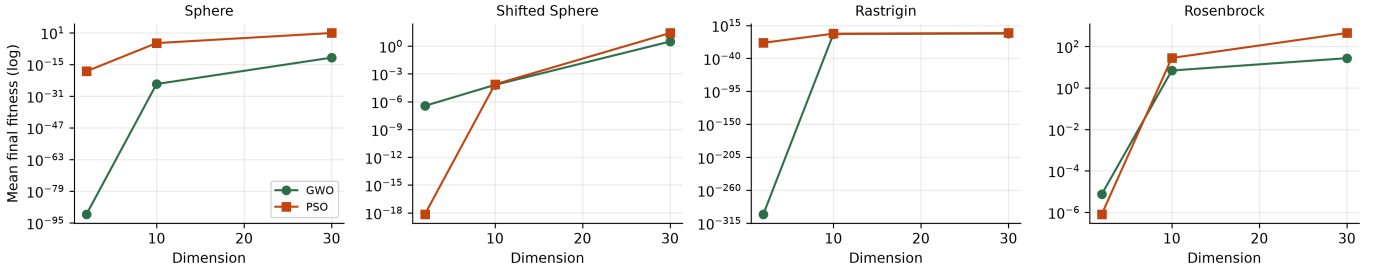


Fig. 5: Scalability of solution quality with problem dimension.

VIII. DISCUSSION

The results support a qualified answer to the research question, under matched population size and iteration budget, GWO produced statistically better final solution quality than the PSO baseline in half (6/12) of tested configurations, with the advantage concentrated at higher problem dimensionality and absent or reversed at low dimensionality. This is consistent with, though should not be over-read as confirming, the original GWO paper’s claim of a favorable exploration-exploitation balance [1]: the leader-averaging mechanism gives every omega wolf’s next position information from three independently perturbed estimates of the best region found so far, which may act as an implicit variance-reduction step in high dimensions. We emphasize three caveats. First, we tested one specific, un-tuned PSO parameterization; numerous refinements exist that could close or reverse the observed gap, so this study does not claim GWO is superior to PSO in general, only to this specific, common baseline. Second, both algorithms were evaluated on only four functions from a much larger benchmark landscape, 175 functions classified by [7]. Third, and most fundamentally, the No Free Lunch theorem [6] guarantees that any comparative advantage measured on a finite function sample cannot be extrapolated to general superiority. The low-dimensional reversals (PSO outperforming GWO on Shifted-Sphere-2D and Rosenbrock-2D) show that GWO’s advantage is not uniform across the design space and that simple, low-dimensional, well-behaved landscapes are exactly the regime in which a momentum-based method like PSO’s velocity update can out-exploit GWO’s discrete leader-averaging step.

IX. LIMITATIONS

- Two algorithms, one baseline. A more complete study would include Differential Evolution, a Genetic Algorithm and/ or covariance matrix adaptation evolution strategy, and a random-search control.
- Four benchmark functions, all are standard functions with no real-engineering-application structure, because external validity is not established.
- Fixed hyperparameters (population 30, iterations 200) across all configurations rather than per-function tuning.
- No hybrid/improved GWO variants tested, only the base algorithm as originally published.

- No multiple-comparisons correction across the 12 Wilcoxon tests, though the smallest significant p -values ($< 10^{-5}$) are well below any reasonable corrected threshold.
- Coursework-scale computational budget (25 trials/configuration on a single machine).

X. CONCLUSION

We presented an independent implementation of the Grey Wolf Optimizer (GWO) against Particle Swarm optimizer (PSO) baseline, and conducted a statistically grounded comparative benchmarking study across four standard continuous test functions, three dimensionalities, and 600 total optimization runs. GWO outperformed the PSO baseline with statistical significance in half of the tested configurations, with the advantage concentrated at higher problem dimensionality. The PSO held a small but significant edge in two low-dimensional configurations, and both algorithms failed to reliably solve the hardest high-dimensional landscapes within the tested budget. We formulate this explicitly as an implementation and replication study with a genuine comparative empirical contribution not a novel algorithmic proposal..

XI. FUTURE WORK

- A natural extension of this would be the development of a novel GWO variant specifically tailored to high-dimensional search, potentially through adaptive parameter control, diversity preservation, or hybrid exploration-exploitation mechanisms.
- Extend the comparator set to Differential Evolution, covariance matrix adaptation evolution strategy, and a random-search control.
- Evaluate on the full [7] catalogue or a congress on evolutionary computation competition benchmark suite (rotated, shifted, composition functions).
- Apply both algorithms to a real, constrained engineering design problem.
- Conduct a hyperparameter sensitivity sweep and report performance profiles rather than single-budget point comparisons.

DATA AND CODE AVAILABILITY

The complete implementation, experimental scripts, and figure-generation codes accompanying this paper can be found here: . The resources provided are intended to facilitate the reproducibility of the experimental results and reported figures.

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